

From these Indian numbers,
0 9 8 7 6 5 4 3 2 1,
we derive great benefit

The Universe is
written in the
language of
mathematics



ERATOSTHENES
276–194 BC

The Greek scholar Eratosthenes was not just good at maths but at astronomy, geography, and history, too. He devised a way of hunting for prime numbers, he drew maps of the known world and the night sky, and he figured out the need for leap years. But best of all he worked out the size of Earth before most people knew it was round. His calculation led him to believe there must be a vast area of uncharted ocean – and he was right.



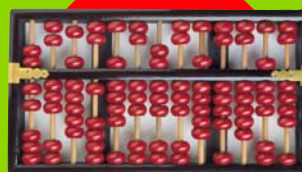
AL KHWARIZMI
780–850 AD

The Arab mathematician Al Khwarizmi lived in Baghdad. He wrote two books about maths that helped spread Indian numbers and zero to the rest of the world. The terms arithmetic and algorithm both come from distortions of his name, and the word algebra comes from the title of his first book, *Ilm al-jabr wa'l muqabalab*. Also a geographer, he helped create a detailed map of the known world.



FIBONACCI
1170–1250

Leonardo da Pisa is best known by his nickname Fibonacci. The son of a travelling Italian merchant, he spent much of his life in Algeria, where the Arabs taught him how to use Indian numbers. Impressed by how these made sums much easier, he wrote a book about them and so made them popular in Italy. He also discovered the Fibonacci series of numbers, which has links to nature and to the golden ratio.



GALILEO
1564–1642

Called the first true scientist, Galileo made telescopes and discovered Jupiter's moons, mountains on the Moon, and sunspots, which eventually blinded him. He also explored the force of gravity. He dropped balls off tall buildings but couldn't time their fall, so he rolled them down slopes instead. He showed they always increase speed in ratio with the square of the time taken. This helped Newton discover gravity.

